Pandas

Description	Library for data manipulation and analysis

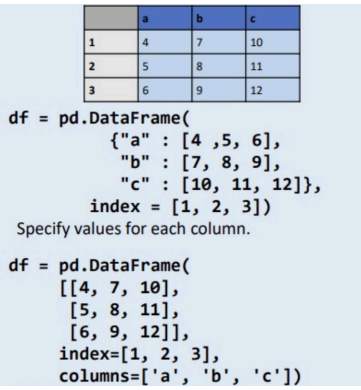
Data Frame

A Data Frame is a two-dimensional Pandas data structure that can be used to organize and manipulate tabular data.

Data extracted from web pages can be loaded into a Data Frame for ***easier manipulation.***

```
import pandas as pd

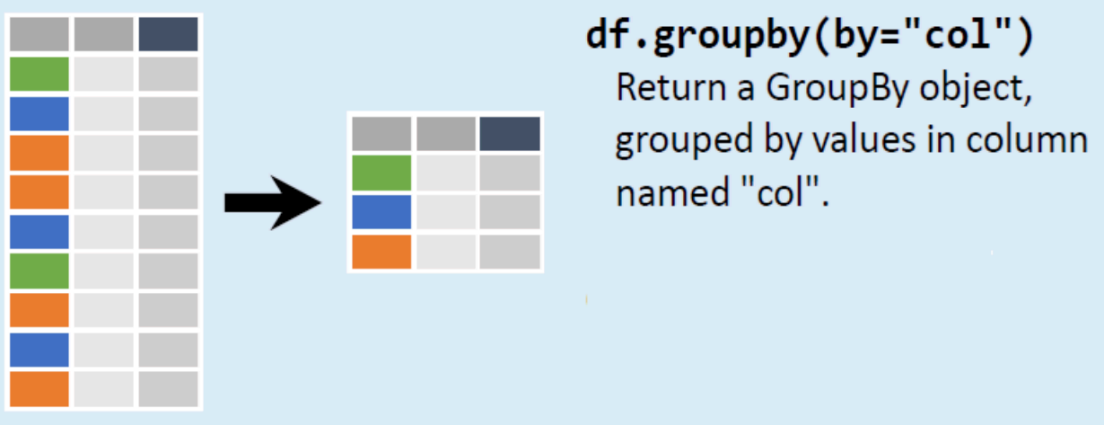
# Creating a DataFrame from extracted data
data = {'Name': ['Alice', 'Bob', 'Charlie'],
        'Age': [25, 30, 35]}
df = pd.DataFrame(data)
```



Data Grouping

Pandas provides data grouping functionalities, useful when you want to aggregate or analyze data based on certain characteristics.

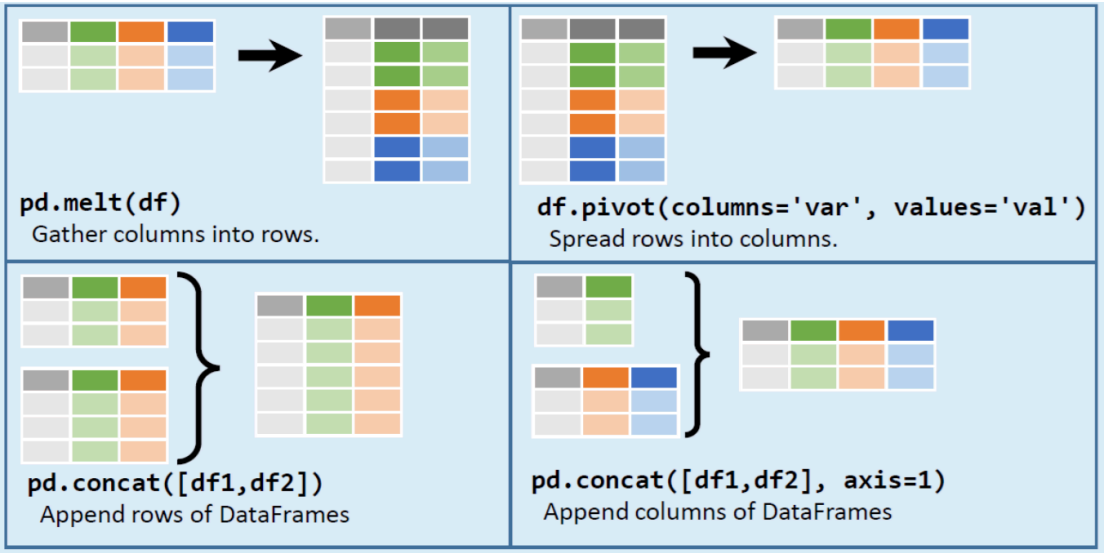
```
# Grouping by age and calculating the mean age for each group
grouped_df = df.groupby('Age').mean()
```



Data Reshaping

Data reshaping is useful when you need to change the shape of the DataFrame. Pandas offers many functions for reshaping, such as transposing and melting data.

```
# Transposing the DataFrame
transposed_df = df.transpose()
```



Data Combining

Pandas allows combining data from different sources. It can be useful when collecting data from various web pages or sources and wanting to combine such data into a single DataFrame.

```
# Combining two DataFrames based on a common column
combined_df = pd.merge(df1, df2, on='common_column')
```

